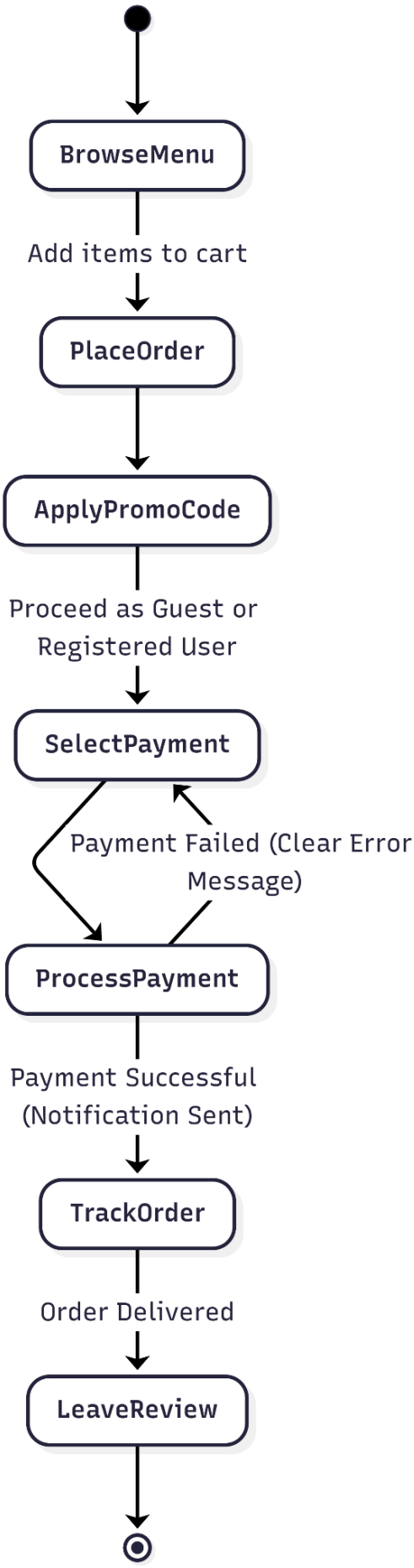
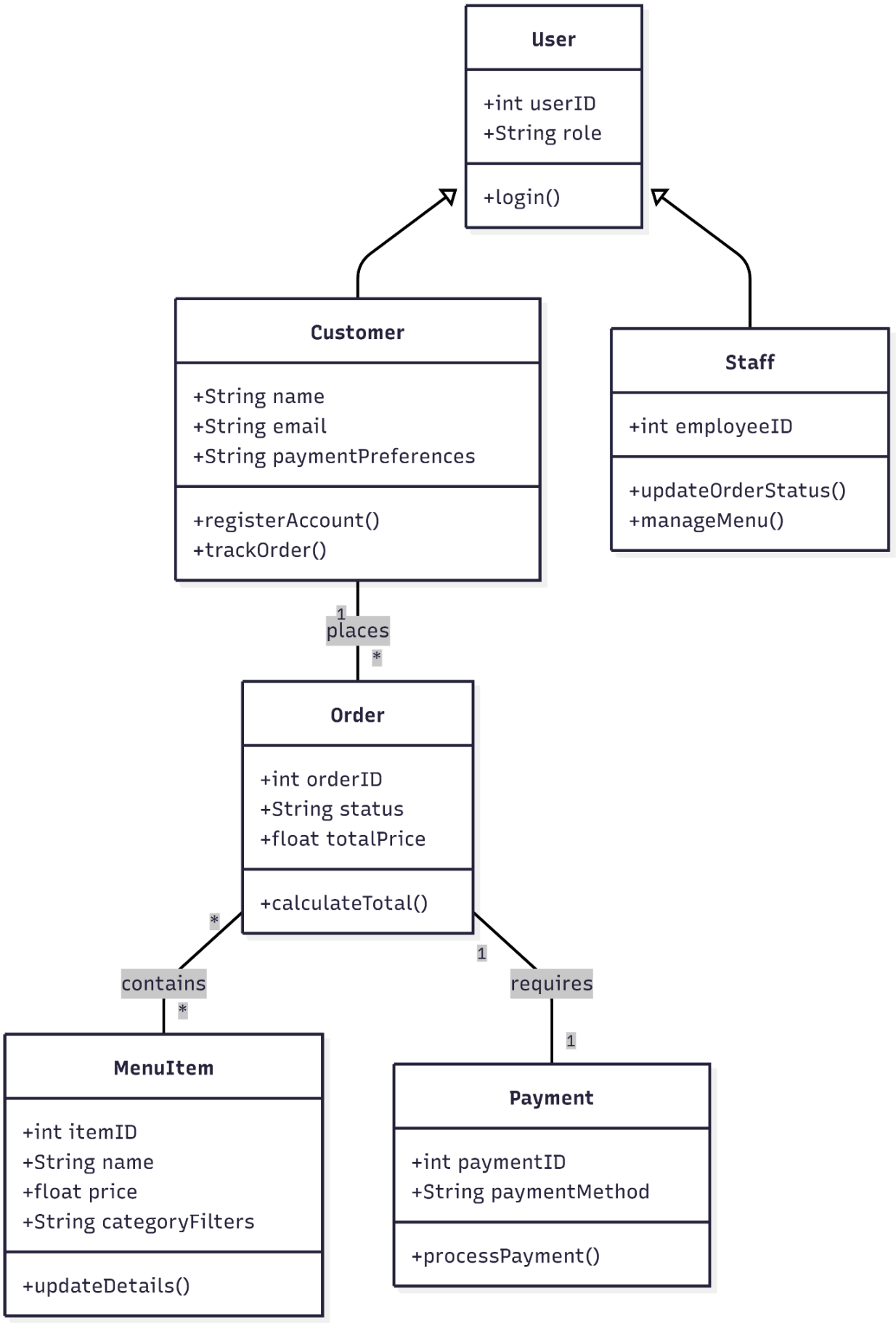
1. **Sequencing Diagram of placing an Order**

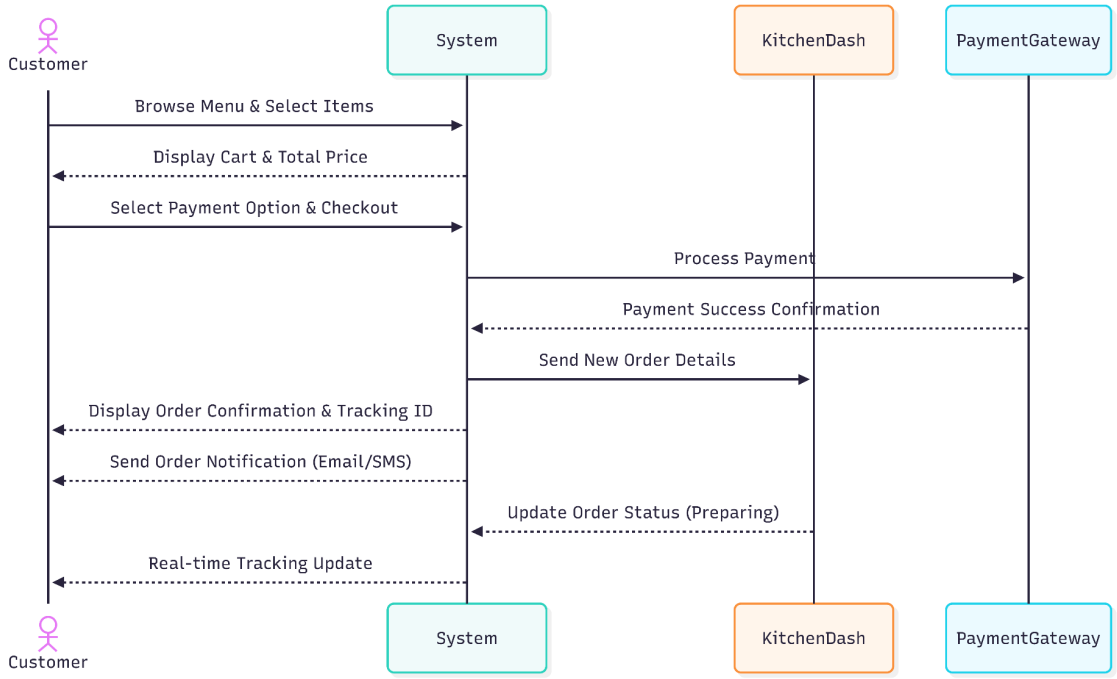
This depicts the step-by-step interactions over time for the high-priority order use case. It also incorporates the multiple payment options and notifications features.


BrowseMenu
Add items to cart
PlaceOrder
ApplyPromoCode
Proceed as Guest or Registered User
SelectPayment
Payment Failed (Clear Error Message)
ProcessPayment
Payment Successful (Notification Sent)
TrackOrder
Order Delivered
LeaveReview

2. **Class Diagram of Database / Schema**

This diagram is optimized for your database design consideration guidelines. It explicitly defines Primary Keys (PK) for every table and shows clear relationships to help build separate model and schema files.


User
+int userID
+String role
+login()
Customer
+String name
+String email
+String paymentPreferences
+registerAccount()
+trackOrder()
Staff
+int employeeID
+updateOrderStatus()
+manageMenu()
1
places
*
Order
+int orderID
+String status
+float totalPrice
+calculateTotal()
*
contains
*
1
requires
1
MenuItem
+int itemID
+String name
+float price
+String categoryFilters
+updateDetails()
Payment
+int paymentID
+String paymentMethod
+processPayment()

3. **Logical Flow diagram of Placing an Order**

This diagram maps the logical sequencing requested, ensuring an order is created and paid for before tracking or reviewing can occur.